From four-leg mass filters to correlated time smearing in dS_3

Alan Day Chengheng*

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Abstract

The Gaussian mass regulator used in the existing static-patch trace argument is not four independent time smearings. We decompose it exactly into a mean-mass window and a correlated Gaussian time convolution. For four legs the relative pair differences retain their variance, while the pair-center variance is halved. We apply this identity to the real-principal eikonal model, derive its small-filter correction, and test a separately stipulated mass-averaging model. This refines the comparison target; it neither constructs a physical finite-G state nor resolves the known trace obstruction.

1 The source quantity and the exact filter identity

Set $\ell = 1$. CSTY (2.3) defines the mixed representation

$$F(\theta_i \mid \mu) = \frac{e^{-2\pi\mu}}{Z_{\text{sphere}}(\mu)} \int d^4 m_i \, \delta\left(\mu - \frac{1}{4}\sum_i m_i\right) e^{\sum_i (\theta_i - \theta_{i-1}) m_i} \boldsymbol{F}(m_i). \tag{1}$$

Its proposed spectral trace identification and four-leg Gaussian regulator occur explicitly in CSTY Section 3.1 [1]. They are prior constructions, not new states introduced here. At equal Euclidean spacings $\pi/2$, the positive mass filter is $s_\delta(m_i) = \exp[-\delta^2(m_i - M)^2]$. Write $\mu = \frac{1}{4}\sum_i m_i$ and $s_i = m_i - \mu$. Then exactly

$$\prod_i s_\delta(m_i) = e^{-4\delta^2(\mu - M)^2} \, e^{-\delta^2 \sum_i s_i^2}. \tag{2}$$

Restricting only μ drops the second factor.

Define the cyclic difference matrix by $(Dm)_i = m_i - m_{i+1}$, with $m_5 = m_1$, and $P = I - \mathbf{1}\mathbf{1}^T/4$. If

$$C = \frac{\delta^2}{8}\begin{pmatrix} 5 & -1 & -3 & -1 \\ -1 & 5 & -1 & -3 \\ -3 & -1 & 5 & -1 \\ -1 & -3 & -1 & 5 \end{pmatrix},$$

then $C = 2\delta^2 (DD^T)^+$ and $D^T C D = 2\delta^2 P$. For a centered Gaussian d of covariance C on $\sum d_i = 0$,

$$\mathbb{E}_d e^{i(Dm)\cdot d} = e^{-\delta^2 \sum_i s_i^2}. \tag{3}$$

The common-time zero mode is immaterial. This is the exact quadratic transfer factor left as an order estimate in the source argument. It is an application of elementary Gaussian Fourier algebra, not a priority claim.

*AI research assistant (OpenAI).

2 Correlations, normalization, and the remaining mass average

Use the same insertion ordering as the preceding real-principal eikonal calculation:

$$\theta = (-\pi/2 + i(t + d_1),\ id_2,\ \pi/2 + i(t + d_3),\ \pi + id_4).$$

The combinations

$$X = (d_1 - d_3)/2, \qquad Y = (d_2 - d_4)/2, \qquad Z = (d_1 + d_3 - d_2 - d_4)/2$$

are independent with

$$\operatorname{Var} X = \operatorname{Var} Y = \operatorname{Var} Z = \delta^2/2. \tag{4}$$

For four independent insertion shifts of variance δ^2, the first two variances agree but $\operatorname{Var} Z = \delta^2$. Thus the two protocols cannot be interchanged.

Denote this correlated convolution by $\mathcal{C}_\delta$. Inserting (3) into the spectral integral gives the unnormalized filtered quantity, under the same spectral trace hypothesis as the source:

$$\mathcal{T}_{M,\delta} = \int d\mu\, e^{2\pi\mu} Z_{\text{sphere}}(\mu) e^{-4\delta^2(\mu - M)^2} \mathcal{C}_\delta F(\theta_i \mid \mu). \tag{5}$$

All spectral support restrictions must be retained in F. Equation (5) is also a formal transform identity if that hypothesis is not assumed. In that case it is not established as a positive trace or a state preparation.

For the specified eikonal model, take real principal labels $\nu, \nu' > 0$, probe masses squared $1 + \nu^2, 1 + \nu'^2$, $\gamma(\mu) = 4G\mu$, and $\alpha(\mu) = 2\pi(1 - \gamma(\mu))$. Let

$$I_{n,\nu}(X) = \int_0^\infty x^{n+1} K_{i\nu}(xe^X) K_{i\nu}(xe^{-X})\, dx, \qquad B_{n,\nu}(\delta) = \mathbb{E}_X I_{n,\nu}(X).$$

The disconnected term is still $D_\delta(\mu) = B_{0,\nu} B_{0,\nu'} / [\pi\alpha(\mu)]^2$. If the numerator in (5) is divided by the corresponding disconnected integral, its normalized version is

$$\overline{\mathcal{R}} = \frac{\int d\mu\, a(\mu) D_\delta(\mu) \mathcal{R}^{\text{corr}}_\delta(t \mid \mu)}{\int d\mu\, a(\mu) D_\delta(\mu)}, \qquad a(\mu) = e^{2\pi\mu} Z_{\text{sphere}}(\mu) e^{-4\delta^2(\mu - M)^2}. \tag{6}$$

It is a positive average only if aD_δ is positive and integrable. The source normalization constant and actual time-ordered factorization remain separate questions. In particular, (6) does not identify an arbitrary $G = 0$ observer wavepacket with the mixed representation. At fixed theory G stays fixed during the μ integral; holding γ fixed would average different theories.

3 A calculable change in the real-principal coefficients

The previous lateral kernel representation remains valid with (4). In detail, put

$$A = e^{Y}\cosh r_1 + e^{-Y}\cosh r_2, \quad B = e^{X}\cosh r_3 + e^{-X}\cosh r_4, \quad T = t + Z,$$

and $W = \cos\nu r_1 \cos\nu r_2 \cos\nu' r_3 \cos\nu' r_4$. For $\eta = +1$ the continued kernel is

$$\mathcal{K}_\eta(c) = -J'_\eta(c), \quad J_\eta(-1/a) = ae^{-a}\,\mathrm{Ei}(a) - \eta i\pi a e^{-a}.$$

The conditional normalized model is

$$\mathcal{R}^{\mathrm{corr}}_\eta = \frac{\mathbb{E}\left\langle \int_{\mathbb{R}^4_+} \frac{W}{A^2B^2}\left[\mathcal{K}_\eta(he^{T}/(AB)) + \mathcal{K}_{-\eta}(he^{-T}/(AB)) - 1\right] d^4r \right\rangle_\phi}{B_{0,\nu}B_{0,\nu'}}.$$

Here $h = -\cos[(1-\gamma)(\pi-\phi)]/[2\mu\,\mathrm{sinc}(\pi\gamma)]$. It retains both phases. At each fixed perturbative order the factor is

$$\frac{(-1)^n\langle h^n\rangle}{n!}e^{n^2\delta^2/4}\frac{B_{n,\nu}B_{n,\nu'}}{B_{0,\nu}B_{0,\nu'}}(e^{nt}+e^{-nt}). \tag{7}$$

The independent-time prescription instead has $e^{n^2\delta^2/2}$. Moment positivity still follows from its established Mellin spectrum; it does not prove positivity of the full gravitational theory.

The small-width change is explicit. With $a = (n+2)/2$ and $f_n(r) = e^{ar}K_{i\nu}(e^r)$, integration by parts in the Bessel equation gives

$$\frac{\|f_n'\|_2^2}{\|f_n\|_2^2} = a^2 + \nu^2 - \frac{I_{n+2,\nu}(0)}{I_{n,\nu}(0)} = q_{n,\nu} := \frac{(n+2)^2/4 + \nu^2}{n+3}.$$

Boundary terms vanish for $n \geq 0$ and real ν. The Mellin transform and Gaussian multiplier then imply

$$B_{n,\nu}(\delta) = I_{n,\nu}(0)[1 - q_{n,\nu}\delta^2 + O(\delta^4)].$$

The coefficient in (7), relative to zero width, changes by $1 + c_n\delta^2 + O(\delta^4)$, where

$$c_n = \frac{n^2}{4} - q_{n,\nu} + q_{0,\nu} - q_{n,\nu'} + q_{0,\nu'}, \qquad c_1 = \frac{2(\nu^2+\nu'^2) - 5}{24}.$$

For $(\nu,\nu') = (.6,.9)$, $c_1 = -.110833\ldots$; independent time smearing gives $+.139167\ldots$. The sign of this width correction differs, while the first gravitational growing term itself still has the positive sign of the existing eikonal model.

4 Endpoint control and an explicitly stipulated comparison

The mass integral is not harmless merely because the Gaussian is small far from M. Combining the large-mass approximation $Z_{\mathrm{sphere}}(\mu) \propto \mu^2 e^{-2\pi\mu}$ with $D_\delta \propto \alpha^{-2}$ over $0 < \mu < \kappa = 1/(4G)$ would give a weight proportional to

$$\frac{\mu^2 e^{-4\delta^2(\mu-M)^2}}{(\kappa-\mu)^2},$$

whose integral diverges at $\mu = \kappa$ for every finite δ. This is failure of an extrapolated approximation, not a physical prediction of divergent trace normalization.

For a concrete finite comparison, we separately stipulate the semiclassical relation $\alpha = 2\pi(1-\mu/\kappa)$ at finite κ and use the CLP proposal $Z_{\mathrm{obs}}(\alpha) = 32\sin^2(\alpha/2)\sinh^2(\kappa\alpha/2)$ [2]. This finite-κ map and the insertion completion are not established. After removing a common positive constant, the weight in (6) is exactly

$$A_\kappa(\mu) e^{-4\delta^2(\mu-M)^2}, \qquad A_\kappa(\mu) = \left[\frac{\sin(\alpha/2)}{\alpha}(1-e^{-\kappa\alpha})\right]^2, \quad 0 < \mu < \kappa. \tag{8}$$

It satisfies $0 \le A_\kappa \le 1/4$ and tends to $\pi^2(\kappa-\mu)^2$ at the upper endpoint. The mass average of the bounded conditional eikonal ratio therefore exists. This restores an integrable comparison weight, not an established physical finite-G density matrix.

On any fixed interior relative mass window, the preceding rapidity representation has bounded second derivatives in $r = \mu/M$ and in the three displacement variables, for fixed principal labels, $\gamma(M)$ away from one and $t = \log M + u$ with u in a compact set. The bound follows by differentiating the smooth real lateral kernel; its derivatives are bounded, Gaussian exponential moments are finite, and the rapidity majorant is integrable. Taylor expansion with centered Gaussians then bounds the conditional smearing error by $O(\delta^2)$ and the normalized mean-mass averaging error by $O((\delta^2 M^2)^{-1})$, provided the positive density has bounded relative C^2 derivatives there. For (8), its endpoint bound supplies an exponentially small exterior probability when $\delta^2 \asymp M^{-1}$ and $\gamma(M)$ is fixed in $(0,1)$. Thus the source's $O(M^{-1})$ balancing estimate holds for this stipulated comparison model. The constants are not bounds on omitted gravitational terms.

What is still missing. CLP (3.35)–(3.37) only calibrates the physical insertion in a leading heavy-heavy-light limit. Its positive two-point expression does not fix the finite-parameter four-point Gram inverse or Lorentzian state. Our filter identity neither supplies these data nor invalidates the source's positivity argument, which already uses the four separate mass filters.

5 Numerical checks and reproducibility

Exact rational algebra checks $D^T C D = 2\delta^2 P$ and the reduced covariance. A three-level Hermitian matrix example compares the actual filtered trace with independent Gaussian quadrature of the loop transform; both agree to the saved tolerance. The mean-only window is also retained and gives a different result. Mellin second moments independently test $q_{n,\nu}$ at 50 decimal digits. All 23 diagnostics pass; this finite matrix check is not a test of quantum-gravity dynamics.

For the conditional eikonal integrals we use $\nu = .6, \nu' = .9$, $\mu = 1000$, $\gamma = .25$, $t = \log 1000 + u$. Eight scrambled Sobol seeds at 2^{19} and 2^{21} points retain both protocols with common samples. The following values are correlated minus independent, with empirical component standard errors across scrambles:

δ	u	$\Re\Delta\mathcal{R}$	$\Im\Delta\mathcal{R}$	$\mathrm{SE}_{\Re}$	$\mathrm{SE}_{\Im}$
$1/\sqrt{1000}$	0	$-.00015810$	.00022730	$5.14\,10^{-8}$	$4.86\,10^{-8}$
.4	0	$-.007832$	.033554	.0000071	.0000067
.8	0	.032164	.079138	.0000189	.0000315
.8	2	$-.079416$	$-.147868$	.0000434	.0000599

These are full conditional kernel integrals, not just perturbative coefficients. They are not mass-integrated traces.

A separate nine-dimensional run implements (8) at $M = 1000$, $\kappa = 4000$, $G = 1/16000$, $\delta = 1/\sqrt{1000}$ and $u = 0$, with antithetic mass offsets. Its complete raw estimates, mass normalizer and paired differences are retained in `results/stage4/mass-filter/packet.json`. The comparison is explicitly labeled as the stipulated CLP-weight model; a change of mass never changes G. Empirical scramble errors are not rigorous total error certificates.

All drivers refuse to overwrite raw evidence. The stage records dependency versions, code/input hashes, finite-matrix controls, and both sampling levels. The next task is to determine a physical finite-parameter insertion and gluing prescription, or to establish a concrete obstruction to one, retaining the exact filter protocol derived here.

References

[1] Y. Chen, D. Stanford, H. Tang and Z. Yang, *Negative shocks versus static patch holography*, arXiv:2607.14042v1, (2.3), Section 3.1 and Appendix F.

[2] J. Chandra, D. Liska and B. Post, *3D de Sitter from complex Virasoro TQFT: Observers and knots*, arXiv:2610.30247v1, (3.5), (3.10), (3.35)–(3.40).